Chapter 30

Therapeutic Drug Monitoring

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Learning Objective:

Objective 30.1 Describe the concept and need for therapeutic drug monitoring.

OBJECTIVE 30.1. DESCRIBE THE CONCEPT AND NEED FOR THERAPEUTIC DRUG MONITORING

Drug Dosing and Personalized Drug Therapy

Dosing of medications can be fixed (also called "flat" dosing) or variable. Fixed dosing refers to the use of a single dose for all patients regardless of any subject-specific variables such as age, body weight, body surface area, or disease conditions. With variable dosing on the other hand, such factors are usually considered and the dose is adjusted for. For example, weight-based dosing means the required dose for a patient depends on the patient's body weight, whereas body surface area–based dosing means the patient's dose depends on the patient's body surface area, which is a better measure of the overall metabolic capacity in a person. Personalized dosing requires considering and adjusting for more variables such as the pharmacodynamics (response-based/logical dosing) and pharmacokinetics of the product. Personalization may be based on the genotype and/or the phenotype of the patient. Genotyping is the process of determining differences in the genetic make-up (genotype) of an individual by examining the individual's DNA sequence using biological assays and comparing it with another individual's sequence or a reference sequence. It reveals the alleles an individual has inherited from their parents. Genotyping has advantages of being an easy test, can help identify starting dose, is useful for several drugs once done, and can be part of an individual's medical record. Limitations of genotype-based dosing include cost, complex algorithms necessary to decipher complex interactions with multiple genotypes, and potential for compensatory pathways that can mask the impact of a given genotype when multiple enzymes are involved. Furthermore, genotyping error can occur, which poses a major risk of lifelong mismanagement of health issues of the patient as genotypes are lifelong test results.

A phenotype (from Greek "phainein" and "typos," meaning "to show" and "type," respectively) is the composite of an organism's observable characteristics or traits such as its morphology, development, biochemical or physiological properties, phenology, behavior, and products of behavior. Phenotypic measure is determined by various pharmacokinetic properties such as absorption, distribution, metabolism, and excretion of a product. In case of drugs that are primarily eliminated by the kidney, the phenotypic surrogate measure that can be used is a measure of glomerular filtration rate (creatinine clearance). Drugs that are metabolized more commonly require monitoring unlike drugs that are primarily renally excreted (without being significantly metabolized). Phenotypic measures take into account genetic and other factors contributing to variability in drug exposure in a patient.

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Optimizing the Use of Medications: Therapeutic Drug Monitoring

Therapeutic drug monitoring (TDM) is a clinical practice that involves measuring concentrations of a drug in a patient's blood or plasma at designated intervals to provide guidance on individualized dosage regimen and adjustment of drug dosage to maintain a suitable concentration of the drug in the patient's circulation.[1] TDM is required to ensure that therapeutic levels for maximal effectiveness and minimal side effects are achieved in each individual patient.

For a drug to produce its desired therapeutic benefit, it must be present in adequate concentration at the site of action. If the drug concentration is too low (ineffective) or too high (toxic), there will be therapeutic failure. The minimum effective concentration (MEC) of a drug is the lowest concentration of the drug required to achieve the therapeutic benefit. On the other hand, the maximum therapeutic concentration or minimum toxic concentration (MTC) is the concentration at which a drug produces unwanted side effects. In between the MEC and the MTC is the concentration range at which the drug produces optimal therapeutic outcome. This is the therapeutic window, illustrated in Fig. 30.1.

The therapeutic range or therapeutic window of a drug is the concentration range in plasma or blood at which the drug has been determined to be efficacious with minimal toxic effects in most people. Depending on the interval between the MEC and MTC, the therapeutic window of a drug may be narrow or wide.

The therapeutic window is the range of concentrations at which the drug is effective with acceptable side effects.

Goal of Therapeutic Drug Monitoring

There are two main goals of TDM. First is to maximize the effectiveness of the drug and therapeutic benefit. The second is to minimize drug toxicity and adverse effects.

Why Is Therapeutic Drug Monitoring Necessary

TDM is helpful to detect and prevent suboptimal treatment and avoid drug toxicity in patients. If a drug is used at a concentration below the lower limit of its therapeutic range, the drug will be ineffective. If used at a concentration above the upper limit of the therapeutic range, the drug will be toxic. Although the therapeutic window is wide for some drugs, some drugs have a narrow therapeutic window and must be monitored to prevent therapeutic failure (undertreatment or toxicity). Consequences of therapeutic failure may be serious. For example, in patients on immunosuppressive drugs following organ transplantation, inappropriate drug concentrations may result in organ rejection or increased susceptibility to infections due to increased immunosuppression.

TDM is useful for individualization of drug dosage and therapy. Due to exogenous and endogenous factors such as variations in drug metabolism, individual patients may respond differently to similar doses of certain drugs with same dose producing variable drug exposure. This interindividual variability necessitates adequate monitoring of such drugs to maximize effectiveness and prevent toxicity.

TDM helps with monitoring and detection of drug and disease-based interactions. In patients with conditions requiring polypharmacy or when changes in a patient's clinical state necessitate changing drugs or adding new drugs, drug−drug interactions need to be monitored to ensure proper adjustment of drug doses to maximize effectiveness. For example, in patients on digitalis therapy, concomitant use of quinidine necessitates adjustment of digoxin dose because quinidine decreases renal clearance of digoxin thereby predisposing the patient to digoxin toxicity. Another example is verapamil which also impairs digoxin clearance, and coadministration requires decreased digoxin dosing.

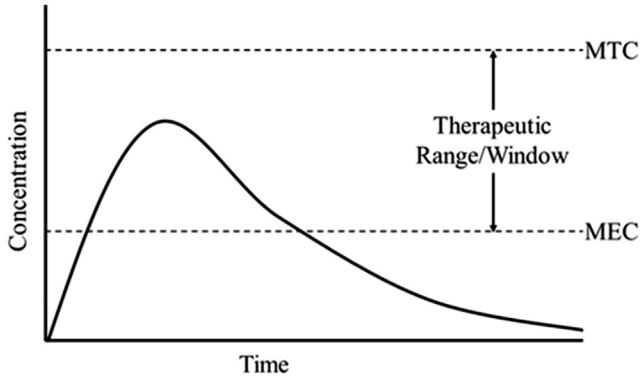


FIGURE 30.1 Plasma concentration Vs time curve. *MEC*, minimum effective concentration; *MTC*, maximum therapeutic concentration.

Furthermore, in practice, TDM is warranted in certain clinical scenarios. Patients who do not respond appropriately to treatment or whose responses raise the suspicion of drug toxicity will require TDM for proper dose adjustments. Similarly, patients on long-term therapy and those with major organ failure (such as hepatic or renal impairment) may also benefit from TDM to prevent toxicity. Other benefits of TDM are that it is a simple method to monitor patient compliance and it can be useful for differentiating adverse effects of drugs from disease symptoms.

Drugs that Require Therapeutic Drug Monitoring

Examples of commonly used drugs that require TDM are described in Table 30.1. Typical characteristics of drugs that require TDM include narrow therapeutic range; significant interindividual pharmacokinetic variability; nonlinear pharmacokinetics (drugs following saturation metabolism); drugs whose therapeutic effect cannot be readily assessed by clinical observation; drugs whose plasma levels (or metabolite plasma levels) correlate directly with the pharmacological or toxic effects; and drugs used for treating life-threatening conditions. On the flip side, drugs that do not require TDM are usually those with a wide therapeutic range; irreversible action (i.e., hit and run drugs such as MAO inhibitors, Omeprazole, Reserpine, Guanethidine); clinically quantifiable pharmacological effects which can be monitored by clinical end points such as blood pressure, heart rate, blood sugar, urine volume, and body temperature; drugs whose plasma levels do not correlate with therapeutic or toxic effects; and drugs used to treat less complicated or non life-threatening diseases.

Measurement of Systemic Drug Levels

The principle of TDM is to estimate the concentration of a drug and/or its metabolites in a patient's body fluids. An ideal method should distinguish between compounds of similar structure, drug versus metabolites, detect small amounts of drug/metabolite(s), be simple enough to use as a routine assay, be unaffected by other drugs administered simultaneously, and is rapid in turnaround time.

Laboratory Techniques Used for Therapeutic Drug Monitoring

TDM assays can be specific or nonspecific. Examples of each are listed in Table 30.2, and comparisons of the assays are shown in Table 30.3. Specific assays separate and detect metabolite and parent drugs, providing better pharmacokinetic parameter estimation of a drug and information on its metabolites. Nonspecific assays typically tend to identify a class of compounds and may provide an overestimation of the actual drug concentrations.

Samples Used for Therapeutic Drug Monitoring

TDM typically involves measurement of drug concentrations in patient's biological fluids. These include the following:

1. Plasma/serum: this is more commonly used as drug concentrations are relatively easy to measure in plasma or serum.
2. Whole blood: more physiologically relevant but less commonly used due to need for generally more technically demanding assay.
3. Saliva: sample collection is noninvasive. Useful for measuring concentration of unbound drug; however, it may be inaccurate due to drug binding to salivary proteins or oral debris. Can be used for detecting drugs such as phenytoin, lithium, and amitriptyline.
4. Breath: Commonly used for detecting alcohol.

Timing of Sample Collection

Sample collection time(s) and the frequency of sampling for TDM vary depending on the pharmacokinetics of the drug being monitored and the purpose of TDM (see Table 30.4). Factors that determine the frequency of sampling include the half-life of the drug; length of drug use; change in dose; change in drugs; change in patient's hepatic or renal function; and clinically mandated factors such as rejection or toxicity.

Clinical Interpretation of Therapeutic Drug Monitoring Reports

If there are alterations in serum drug concentration, either lower or higher than expected, or if a patient fails to respond to therapy despite adequate serum drug concentration, then drug dosage adjustment is required. The common causes of alterations in serum drug concentration in clinical practice are listed in Table 30.5.

TABLE 30.1 Examples of Drugs That Require Therapeutic Drug Monitoring (TDM)

Drug	Biological Matrix	Assay Method[a]	Therapeutic Range	Consequences of No TDM	When to Measure[c]	What to Measure
Antiepileptics[2]						
Carbamazepine	Serum Plasma	LC-UV LC-MS GC-TSD FPIA EMIT MECC	4–12 mg/L[b]	↓Lack of therapeutic response ↑Lethargy, dizziness, headache, blurred vision, nystagmus, diplopia, ataxia, incoordination, sedation, nausea, vomiting, cardiac disturbances, osteomalacia, etc.	Trough level	Parent drug and active metabolite: carbamazepine-10,11-epoxide
Phenytoin	Serum Plasma	FPIA EMIT GC-TSD LC-UV MECC	10–20 mg/L	↓Lack of therapeutic response ↑Confusion, ataxia, coma, nystagmus, diplopia, cognitive and behavioral dysfunctions, exacerbation of seizure frequency, folate and vitamin D deficiency, etc.	Trough level	Parent drug
Phenobarbital	Serum Plasma	RIA FPIA EMIT GC-TSD LC-UV MECC	10–40 mg/L	↓Lack of therapeutic response ↑Sedation, drowsiness, slowness, ataxia, coma, nystagmus, diplopia, cognitive and behavioral dysfunctions, exacerbation of seizure frequency, folate and vitamin D deficiency, etc.	Trough level	Parent drug
Valproic acid	Serum Plasma	FPIA EMIT GC-FID LC-FLD LC-UV	50–100 mg/L	↓Lack of therapeutic response ↑Nausea, vomiting, diarrhea, sedation, stupor, tremor, malaise, thrombocytopenia, hepatotoxicity, platelet aggregation defects, etc.	Trough level	Parent drug
Topiramate	Serum Plasma	GC-MS GC-FID GC-NPD FPIA EMIT	5–20 mg/L	↓Lack of therapeutic response ↑Psychomotor slowing, trouble concentrating, speech and language difficulties, somnolence, fatigue, etc.	Trough level	Parent drug
Primidone	Serum Plasma	GC-TSD GC-MS FPIA EMIT LC-UV MECC	5–10 mg/L[d]	↓Lack of therapeutic response ↑Sedation, drowsiness, slowness, ataxia, coma, nystagmus, diplopia, cognitive and behavioral dysfunction, exacerbation of seizure frequency, folate and vitamin D deficiency, etc.	Trough level	Parent drug and active metabolite: phenobarbital
Lamotrigine	Serum Plasma	LC-UV EMIT	2.5–15 mg/L	↓Lack of therapeutic response ↑Somnolence, dizziness, sedation, diplopia, skin rash, Stevens-Johnson reaction, toxic epidermis necrolysis, etc.	Trough level	Parent drug

Oxcarbazepine	Serum Plasma	LC-UV GC-MS EMIT	3–35 mg/L[e]	↓Lack of therapeutic response ↑Psychomotor slowing, trouble concentrating, speech and language difficulties, dizziness, somnolence, diplopia, fatigue, nausea, vomiting, ataxia, abnormal vision, gait disturbances abdominal pain, tremor, dyspepsia, abnormal gait, etc.	Trough level	Active metabolite: eslicarbazepine
Levetiracetam	Serum Plasma	LC-UV GC-NPD EMIT	12–46 mg/L	↓Lack of therapeutic response ↑Drowsiness, somnolence, dizziness, fatigue, asthenia, ataxia, abnormal gait or incoordination, agitation, hostility, anxiety, apathy, emotional lability, depersonalization, depression, etc.	Trough level	Parent drug
Antibiotics						
Gentamicin[3]	Serum Plasma	FPIA EMIT	25–30 mg/L (peak)[f] <1 mg/L (trough)[f]	↓Lack of therapeutic response ↑Nephrotoxicity, ototoxicity (dizziness, vertigo, tinnitus, roaring in the ears, hearing loss), neuromuscular blockade, etc.	Peak level Trough level	Parent drug
Amikacin[4]	Serum Plasma	GC-MS FPIA EMIT	40–60 mg/L (peak)[f] <5 mg/L[f]	↓Lack of therapeutic response ↑Nephrotoxicity, ototoxicity (dizziness, vertigo, tinnitus, roaring in the ears, hearing loss), neuromuscular blockade, etc.	Peak level Trough level	Parent drug
Tobramycin[3]	Serum Plasma	FPIA EMIT	25–30 mg/L (peak)[f] <1 mg/L (trough)[f]	↓Lack of therapeutic response ↑Nephrotoxicity, ototoxicity (dizziness, vertigo, tinnitus, roaring in the ears, hearing loss), neuromuscular blockade, etc.	Peak level Trough level	Parent drug
Vancomycin[5]	Serum Plasma	LC-UV LC-FLD LC-MS FPIA EMIT	15–20 mg/L	↓Lack of therapeutic response ↑Nephrotoxicity, ototoxicity, skin rashes and anaphylaxis, red-neck syndrome, chills and fever; phlebitis and pain at site of injection, etc.	Trough level	Parent drug
Immunosuppressants[6]						
Tacrolimus	Whole blood	HPLC-MS MEIA ELISA FPIA EMIT ECLIA	5–15 ng/mL	↓Organ rejection ↑Nephrotoxicity, CNS toxicity (headache, seizure, tremors, altered mental status and motor function), hyperglycemia, hypertension, infections, hyperlipidemia, electrolyte abnormalities, anemia, diarrhea, nausea, pruritus, alopecia, etc.	Trough level[g]	Parent drug

Continued

TABLE 30.1 Examples of Drugs That Require Therapeutic Drug Monitoring (TDM)—cont'd

Drug	Biological Matrix	Assay Method[a]	Therapeutic Range	Consequences of No TDM	When to Measure[c]	What to Measure
Sirolimus[7]	Whole blood	LC-UV LC-MS MEIA Competitive CLIA	5–10 ng/mL	↓Organ rejection ↑Infection, thrombocytopenia, leukopenia, hypertriglyceridemia, etc.	Trough level[g]	Parent drug
Everolimus[8]	Whole blood	LC-UV LC-MS ELISA FPIA	3–8 μg/L	↓Organ rejection ↑Renal dysfunction, infections, thrombocytopenia, hyperlipidemia, hypercholesterolemia, leukopenia, neurological disorder, anemia, gastrointestinal disorder, etc.	Trough level[g]	Parent drug
Mycophenolate mofetil (MMF)	Serum Plasma	LC-MS EMIT	MPA AUC:30–60 mg h/L MPA trough level: 1–3 μg/mL	↓Organ rejection ↑Infection	AUC[g] Trough level[g]	Parent drug MMF and metabolites: MPA and MPAG
Cyclosporine	Whole blood	LC-MS FPIA EMIT RIA ECLIA	100–400 ng/mL	↓Organ rejection ↑Nephrotoxicity, neurotoxicity (tremors and seizures), hypertension, electrolyte abnormalities, infections, hyperlipidemia, gingival hyperplasia, hirsutism, hepatotoxicity, etc.	Trough level[g]	Parent drug
Antipsychotics						
Clozapine	Serum Plasma	LC-MS GC-MS LC-UV RIA	350–600 ng/mL	↓Lack of therapeutic response ↑Confusion, delirium, seizure, obsessive/compulsive symptoms, etc.	Trough level	Parent drug
Lithium[9]	Serum	ISE RSM SM	0.8–1.2 mEq/L for acute mania 0.6–0.8 mEq/L for prophylaxis of mania and/or depression	↓Lack of therapeutic response ↑Tremor, confusion, sedation, slurred speech, vomiting, lethargy, nausea, diarrhea, muscle weakness, polyuria, polydipsia, seizure, hyperreflexia, cardiovascular collapse, etc.	Trough level	Parent drug
Antifungals[10]						
Voriconazole	Serum Plasma	LC-UV LC-MS LC-FLD EMIT	For prophylaxis, >0.5 μg/mL; For therapy >1–2 μg/mL	↓Lack of therapeutic response ↑Severe hepatopathy, unexplained neurological symptoms, etc.	Trough level	Parent drug and its metabolites voriconazole N-oxide
Posaconazole	Serum Plasma	LC-UV LC-MS LC-FLD	For prophylaxis, >0.5 μg/mL; For therapy, >0.5–1.5 μg/mL	↓Lack of therapeutic response	Trough level	Parent drug

Itraconazole	Serum Plasma	LC-UV LC-MS LC-FLD	For prophylaxis, >0.5 μg/mL; For therapy, >1−2 μg/mL	↓Lack of therapeutic response	Trough level	Parent drug
Antineoplastics						
Methotrexate	Serum Plasma	LC-UV EMIT FPIA	<10 μM[h] <1 μM[h] <0.1 μM[h] <0.05 μM[h]	↓Lack of therapeutic response ↑Nausea vomiting, mucositis, rash, hepatotoxicity, leukoencephalopathy, myelosuppression, nephrotoxicity, etc.	20 h postinfusion 44 h postinfusion 68 h postinfusion 92 h postinfusion	Parent drug
Busulfan	Serum Plasma	LC-MS LC-FLD GC-MS GC-ECD	1000−1400 μmol/L·min for 6 h dosing interval; 4000−5600 μmol/L·min for 24 h dosing interval	↓Engraftment failure ↑Hepatic veno-occlusive, central nervous system toxicity, interstitial pneumonia, mucositis, etc.	AUC	Parent drug
Antiarrhythmics						
Digoxin	Serum Plasma	MEIA LC-MS FPIA ECLIA EMIT	0.8−2 μg/L	↓Lack of therapeutic response ↑Decreased heart rate, ECG change, arrhythmias, anorexia, nausea, vomiting, diarrhea, visual disturbances, etc.	Trough level	Parent drug
Lidocaine	Serum Plasma	LC-UV LC-MS GC-MS GC-NPD FPIA EIA	1.5−6 mg/L	↓Lack of therapeutic response ↑Confusion, dizziness, slurred speech, diplopia, tremor, severe nausea, vomiting, seizure, obtundation, hypotension, respiratory depression, sinus bradycardia, sinus arrest, atrioventricular conduction disturbances, etc.	Trough level	Parent drug and active metabolites: monoethylglycinexylidide and glycinexylidide
Antipyretic						
Acetaminophen	Serum Plasma	LC-MS GC-MS EMIT FPIA	10−20 mg/L	↓Lack of therapeutic response ↑Skin rash, drug fever, mucosal lesions, hepatic necrosis, renal tubular necrosis, hypoglycemic coma	Peak level (1 h after a dose)	Parent drug
Antiinflammatory						
Salicylate	Serum Plasma	LC-UV LC-MS GC-MS FPIA	150−300 mg/L	↓Lack of therapeutic response ↑Tinnitus, headache, vertigo, hearing loss, central hyperventilation, respiratory alkalosis, metabolic acidosis, hyperglycemia, hypoglycemia	Peak level (1−3 h after an oral dose)	Salicylic acid

Continued

TABLE 30.1 Examples of Drugs That Require Therapeutic Drug Monitoring (TDM)—cont'd

Drug	Biological Matrix	Assay Method[a]	Therapeutic Range	Consequences of No TDM	When to Measure[c]	What to Measure
Bronchodilator						
Theophylline[11]	Serum Plasma	GC-MS FPIA EMIT CEDIA PETINIA	Asthma: 10–20 mg/L COPD: 5–15 mg/L Apnea or bradycardia in neonates: 5–10 mg/L Ventilator weaning of neonates: 5–20 mg/L	↓Lack of therapeutic response ↑Nausea, vomiting, diarrhea, headache, irritability, nervousness, insomnia, tremor, hyperglycemia, hyperkalemia, hypotension, cardiac arrhythmias, cerebral hypoxia, hyperthermia, seizures, brain damage and death, etc.	Peak level	Parent drug

Key: ↓ correlates to subtherapeutic levels, ↑ correlates to supratherapeutic levels. *AUC*, area under the blood/plasma concentration versus time curve; *CEDIA*, cloned enzyme donor immunoassay; *COPD*, chronic obstructive pulmonary disease; *ECD*, electron capture detection; *ECLIA*, electrochemiluminescence immunoassay; *EMIT*, enzyme multiplied immunoassay; *FID*, flame ionization detector; *FLD*, fluorescence detector; *FPIA*, fluorescence polarization immunoassay; *GC*, gas chromatography; *HPLC*, high-performance liquid chromatography; *ISE*, ion-selective electrode; *MECC*, micellar electrokinetic capillary chromatography; *MEIA*, microparticulate enzyme immunoassay; *MPA*, mycophenolic acid; *MPAG*, mycophenolic acid glucuronide; *MS*, mass spectrometry; *NPD*, nitrogen–phosphorus detector; *PETINIA*, particle enhanced turbidimetric inhibition immunoassay; *RIA*, radioimmunoassay; *RSM*, reflectance spectrophotometry; *SM*, spectrophotometry; *TSD*, thermionic specific detection; *UV*, ultraviolet absorbance detection.

[a]*Immunoassays typically tend to identify a class of compounds and overestimate concentrations. Specific Assays, such as GC and HPLC/UPLC, can separate and detect metabolite and drugs, which can provide better pharmacokinetic parameter estimation and also get information on the metabolites.*

[b]*Active 10,11 epoxide metabolite contributes to clinical effects.*

[c]*Trough level was usually sampled immediately before next dose, peak level was usually sampled at the end of infusion or 1 h after IM for injection.*

[d]*Represents the concentration of Primidone; its active metabolite phenobarbital also contributes largely to clinical effects and its concentrations should also be monitored.*

[e]*Represents the concentration of active monohydroxy-derivative metabolite, eslicarbazepine.*

[f]*Represents the concentrations of once-daily aminoglycosides dosing regimen.*

[g]*TDM should be done daily after the first week post transplantation, 3–4 times per week in the early transplant period, and every 6 months once stable. Therapeutic ranges may differ in different transplant centers, type of organ transplanted, and different period of post transplantation. In transplantation, patients are maintained at the higher end of the therapeutic range initially, and then the desired concentration is often lowered over time to minimize nephrotoxicity and overimmunosuppression.*

[h]*For high doses of methotrexate (HDMTX) and intermediate dose methotrexate (MTX) therapy greater than and equal to 100 mg/m^2, serum concentration monitoring is necessary. Patients with elevated MTX concentration require increased leucovorin doses administered for a sufficient period for the MTX concentration to fall below 0.05 μM.*

TABLE 30.2 Therapeutic Drug Monitoring Assays

Nonspecific Assays	Specific Assays
Immunoassays (radio/enzyme): • Radioimmunoassay (RIA) • Enzyme-linked immunosorbent assay (ELISA) • Enzyme Multiplied Immunotechnique (EMIT) • Fluorescence polarization immunoassay (FPIA) • Nephelometric inhibition immunoassay (NIIA) • Particle-enhanced turbidimetric inhibition immunoassay (PETINIA)	Chromatography (gas/liquid): • Gas Chromatography (EC, FID, NPD, MS) • HPLC-UV • UPLC-UV • HPTLC-UV • UPLC-Mass spectrophotometry • HPLC-Fluorescence spectrometry • HPLC-Mass spectrometry

EC, electron capture detection; *FID*, flame ionization detection; *HPLC*, high-performance liquid chromatography; *HPTLC*, high-performance thin-layer chromatography; *MS*, mass spectrometry; *NPD*, nitrogen–phosphorus detection; *UPLC*, ultra performance liquid chromatography; *UV*, ultraviolet.

TABLE 30.3 Comparison of Therapeutic Drug Monitoring Assays

Immunoassay
• Lowest complexity test (convenient for smaller hospitals) • Least expensive in terms of capital equipment • Nonspecific; high variability between different sample batches and also within same run • More expensive kits (primary cost) • Typically for measuring one drug at a time
HPLC
• Generally specific • Only cost-effective if test volumes are low • Relatively slow • Technically demanding
LC-MS/MS—UPLC-MS/MS
• Highest expense in capital equipment • Can analyze several drugs and metabolites simultaneously • Much less variability than immunoassay or HPLC • Direct reagent costs considerably lower than immunoassay • Technically demanding

HPLC, high-performance liquid chromatography; *LC-MS*, liquid chromatography–mass spectrometry; *LC-MS/MS*, liquid chromatography–tandem mass spectrometry; *UPLC-MS/MS*, ultra-performance liquid chromatography tandem–mass spectrometry.

TABLE 30.4 Guidelines for Sampling Time

Drug Type/Purpose	When to Sample
Drug with short half-life	Sample before dose Obtain samples for measurement of peak and trough concentration levels (Fig. 30.2)
Drug with long half-life	Time point in the dosing interval that is correlated with exposure or area under the plasma concentration versus time curve, typically trough concentrations Obtain single sample
Drug with low clearance	Just before next dose
For assessment of drug toxicity	Any time
For adjustment of dose	Just before next dose
To diagnose suboptimal therapy	Trough level
To detect noncompliance	Trough level

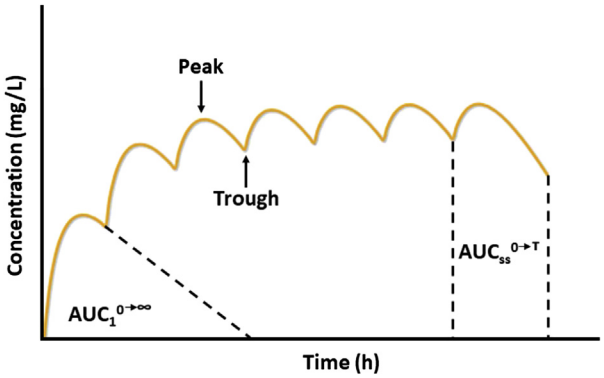


FIGURE 30.2 Plasma concentration Vs time curve. *AUC*, area under the curve - is a measure of drug exposure.

TABLE 30.5 Common Causes of Alterations in Serum Drug Concentration

Drug Concentration Lower Than Expected in Serum
• Noncompliance • Dosing error • Decreased bioavailability due to drug or disease conditions • Increased clearance due to metabolizing enzyme induction due to another drug or dietary supplement • Increased clearance of renally eliminated drug due to physiological changes
Drug Concentration Higher Than Expected in Serum
• Dosing error • Increased bioavailability • Reduced renal clearance • Metabolizing enzyme inhibition • Poor hepatic or renal function
Patient Fails to Respond to Therapy Despite Adequate Drug Concentration in Serum
• Tolerance (impaired receptor sensitivity) • Drug interactions • Change in pathophysiology

Limitations of Therapeutic Drug Monitoring

TDM has shortcomings such as inconsistencies in drug assays used including specificity, sensitivity, accuracy, precision, reproducibility, and limit of detection/quantification. Other limitations of TDM include variations in laboratory reporting methods of assay results; inconsistencies in validity of suggested target ranges depending on other drugs used; limited accessibility to equipments; limited trained/skilled personnel; and high costs.

CONCLUSION

TDM is crucial in clinical practice, to maximize drug effectiveness and minimize toxicity in each individual patient, typically for drugs with narrow therapeutic index.

Case Studies

Case 1: Adjustment of Drug Dosage in a Patient at Risk of Drug Toxicity Due to Drug–Drug Interactions
Source: Jain et al.[12]

Scenario
Mr. A is HIV positive and has been taking the antiretroviral drug Kaletra (combination of lopinavir 133.33 mg and ritonavir 33.3 mg). He takes three capsules of Kaletra twice per day and has undetectable HIV RNA (<50 copies/mL). He was diagnosed with end-stage liver disease, requiring liver transplantation.

Case Studies—cont'd

Therapeutic Drug Monitoring and Clinical Management

Mr. A's Kaletra was temporarily discontinued and, following liver transplantation, he was started on oral Tacrolimus 5 mg twice daily (70 mg weekly). This achieved a 12-h trough Tacrolimus blood concentration of 10.6 ng/mL (therapeutic range, 9.5—11.9 ng/mL). Two weeks later, his Kaletra was restarted. Two days later, Mr. A developed neurologic symptoms including altered mental status. When checked, his trough Tacrolimus blood concentration had increased to 78.5 ng/mL (toxic). Tacrolimus was discontinued. Over the next few days, his neurologic symptoms subsided and his Tacrolimus concentration had dropped to 6.4 ng/mL (subtherapeutic) by 22 days later. Tacrolimus was restarted at 0.5 mg weekly (140× reduction in dose), achieving a mean daily Tacrolimus blood concentration of 12.9 ng/mL (range, 8.9—19.7 ng/mL) while Mr. A continued both medications. His HIV RNA remained undetectable.

Case Discussion

Mr. A needed antiretroviral therapy (Kaletra) to reduce his HIV viral load, as well as immunosuppressant (Tacrolimus) to minimize his risk of organ rejection post transplantation. Tacrolimus has a narrow therapeutic window, thus requires frequent monitoring of its concentration to optimize therapy. Low Tacrolimus concentrations can induce organ rejection, whereas high concentrations can lead to toxicity and immunosuppression-related complications, such as infection. In a patient like Mr. A, who already has immunosuppression from HIV and subsequent increased susceptibility to opportunistic infections, Tacrolimus dosing needs to be appropriately titrated to prevent overimmunosuppression. In this case, Mr. A's concurrent use of Kaletra necessitated a 140-times reduction in his Tacrolimus dose (reduced from 70 mg weekly to 0.5 mg weekly) to achieve a suitable therapeutic level of Tacrolimus.

Case 2: Therapeutic Drug Monitoring-Based Dose Management of Amiodarone When Coadministered With Rifampicin

Source: Oude Munnink et al.[13]

Scenario

Mr. B, a 69-year-old man, has been on amiodarone (AM) (200 mg/day) for about a year, to control arrhythmogenic right ventricular cardiomyopathy. He had comorbid chronic left knee arthritis for which he had knee replacement surgery. His orthopedic prosthesis became infected; hence, Mr. B is initiated on therapy with rifampicin (900 mg/day).

Therapeutic Drug Monitoring and Clinical Management

The potential for drug—drug interaction (DDI) between AM and rifampicin is anticipated; however, no alternate drugs were found suitable for Mr. B. Monitoring of AM and its active metabolite desmethylamiodarone (DAM) was indispensable for continuation of therapy. At baseline, the cumulative AM + DAM concentration was 1.72 mg/L (ref range: 1—4 mg/L) with a metabolic ratio (DAM/AM) of 0.8. Suddenly, 10 days after initiating rifampicin, the cumulative AM + DAM was found to be 0.83 mg/L (reduced by 52%) with metabolic ratio of 1.6. To avoid risk of arrhythmias due to subtherapeutic concentration, AM dose was increased to 600 mg/day. Yet, 1 week later the cumulative AM + DAM concentration was 0.65 mg/L (reduced by 62% from baseline despite threefold dose increase). Consequently, AM dose was further increased to 900 mg/day resulting in therapeutic cumulative AM + DAM concentration of 1.52 mg/L and metabolic ratio <1.

Case Discussion

Two challenges existed in predicting DDI between AM and rifampicin: (1) Induction of Cytochrome P 450 (CYP) enzymes by rifampicin varies with the CYP3A4 effect delayed more than CYP2C8. AM is metabolized by both CYPs, whereas metabolism of DAM is CYP3A4 dependent. (2) Furthermore, AM and DAM have very long half-lives of 50 and 60 days, respectively. Hence, drug monitoring for AM in presence of rifampicin was vital for a timely and appropriate increase in dose. The total duration of coadministration in Mr. B was 90 days, and during this period AM concentrations were maintained within therapeutic range by therapeutic monitoring. Mr. B did not experience any event of arrhythmias.